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United States Patent	12408536
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kim; Jeong-Oh et al.

Display apparatus having a substrate hole and method for forming the same

Abstract

A display apparatus is provided. The display apparatus may include a substrate hole penetrating a device substrate. A light-emitting device may be disposed on the device substrate. The light-emitting device may be spaced away from the substrate hole. A separating device may be disposed between the substrate hole and the light-emitting device. The device substrate may be in contact with the device substrate. Thus, in the display apparatus, the permeation of the moisture through the crack of the insulating layer which occurs in a process of forming the substrate hole may be prevented.

Inventors:	Kim; Jeong-Oh (Goyang-si, KR), Lee; Jei-Hyun (Goyang-si, KR)
Applicant:	LG Display Co., Ltd. (Seoul, KR)
Family ID:	76547478
Assignee:	LG Display Co., Ltd. (Seoul, KR)
Appl. No.:	17/135450
Filed:	December 28, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210202569 A1	Jul. 01, 2021

Foreign Application Priority Data

KR	10-2019-0180185	Dec. 31, 2019
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Publication Classification

Int. Cl.: H10K59/65 (20230101); H10K59/122 (20230101); H10K59/123 (20230101);
H10K59/131 (20230101); H10K59/80 (20230101)

U.S. Cl.:

CPC H10K59/65 (20230201); H10K59/122 (20230201); H10K59/123 (20230201);
H10K59/131 (20230201); H10K59/8723 (20230201); H10K59/873 (20230201)

Field of Classification Search

CPC: H01L (51/5253); H10K (59/873); H10K (50/844)

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Primary Examiner: Bell; Lauren R

Attorney, Agent or Firm: Fenwick & West LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the priority benefit of Republic of Korea Patent Application No. 10-2019-0180185, filed on Dec. 31, 2019, which is hereby incorporated by reference as if fully set forth herein.

BACKGROUND

Field

(2) The present disclosure relates to a display apparatus including a substrate hole which penetrates a device substrate.

Discussion of the Related Art

(3) Generally, an electronic appliance, such as a monitor, a TV, a laptop computer and a digital camera, includes a display apparatus to realize an image. For example, the display apparatus may include a light-emitting device. The light-emitting device may emit light displaying a specific color. For example, the light-emitting device may include a light-emitting layer between a first electrode and a second electrode.

(4) The display apparatus may include a substrate hole in which a peripheral appliance such as a camera, a speaker and a sensor is mounted. The substrate hole may penetrate a device substrate supporting the light-emitting device. The light-emitting device may be spaced away from the substrate hole. A separating device may be disposed between the substrate hole and the light-emitting device. The separating device may cut off an organic layer between the substrate hole and the light-emitting device, partially. For example, the separating device may include an under-cut. The under-cut of the separating device may be formed by the difference of etching ratio between silicon oxide and silicon nitride.

(5) However, in the display apparatus, a crack may be formed in an inorganic insulating layer by a process of forming the substrate hole. Thus, in the display apparatus, external moisture may be permeated through the crack of the inorganic insulating layer.

SUMMARY

(6) Accordingly, the present disclosure is directed to a display apparatus that substantially obviates one or more problems due to limitations and disadvantages of the related art.

(7) An object of the present disclosure is to provide a display apparatus capable of preventing the permeation of the external moisture through the crack which occurs in a process of forming the substrate hole.

- (8) Additional advantages, objects, and features of the invention will be set forth in part in the description which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from practice of the invention. The objectives and other advantages of the disclosure may be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.
- (9) To achieve these objects and other advantages and in accordance with the purpose of the disclosure, as embodied and broadly described herein, there is provided a display apparatus comprising a device substrate. A substrate hole penetrates the device substrate. A light-emitting device is disposed on the device substrate. The light-emitting device is spaced away from the substrate hole. A separating device is disposed between the substrate hole and the light-emitting device. The separating device is in contact with the device substrate.
- (10) The device substrate may include a recess groove. The separating device may be disposed adjacent the recess groove.
- (11) The separating device may include a metal pattern and an insulating cap. The insulating cap may be disposed on the metal pattern.
- (12) The metal pattern may have a smaller width than the insulating cap.
- (13) A dam may be disposed between the light-emitting device and the separating device. At least one insulating layer may be disposed between the device substrate and the dam.
- (14) The dam may have the same stacked structure as the separating device.
- (15) An inclined surface of the insulating layer may have a gentler slope than a side surface of the dam and a side surface of the separating device between the dam and the separating device.
- (16) A driving circuit may be disposed between the device substrate and the light-emitting device. A connecting electrode connecting may be disposed between the driving circuit and the light-emitting device. The light-emitting device may be connected to the driving circuit by the connecting electrode. The metal pattern may include the same material as the connecting electrode.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate embodiment(s) of the invention and together with the description serve to explain the principle of the disclosure.
- (2) FIG. 1 is a view schematically showing a display apparatus according to an embodiment of the present disclosure.
- (3) FIG. 2 is an enlarged view showing a peripheral of a substrate hole in the display apparatus according to an embodiment of the present disclosure.
- (4) FIG. 3A is a view showing a cross-section of a pixel in the display apparatus according to an embodiment of the present disclosure.
- (5) FIG. 3B is a view showing a cross-section of a peripheral region of the substrate hole in the display apparatus according to an embodiment of the present disclosure.
- (6) FIG. 4 is an enlarged view of K in FIG. 3B according to an embodiment of the present disclosure.
- (7) FIGS. 5A, 6A, 7A, 8A, 9A, and 10A and FIGS. 5B, 6B, 7B, 8B, 9B, and 10B are views sequentially showing a method of forming the display apparatus according to an embodiment of the present disclosure.
- (8) FIGS. 11A and 11B are views showing the display apparatus according to another embodiment of the present disclosure.
- (9) FIG. 12 is an enlarged view of P in FIG. 11B according to an embodiment of the present disclosure.

(10) FIGS. 13A and 13B are views sequentially showing a process of forming the display apparatus according to another embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(11) Hereinafter, details related to the above objects, technical configurations, and operational effects of the embodiments of the present disclosure will be clearly understood by the following detailed description with reference to the drawings, which illustrate some embodiments of the present disclosure. Here, the embodiments of the present disclosure are provided in order to allow the technical spirit of the present disclosure to be satisfactorily transferred to those skilled in the art, and thus the present disclosure may be embodied in other forms and is not limited to the embodiments described below.

(12) In addition, the same or extremely similar elements may be designated by the same reference numerals throughout the specification, and in the drawings, the lengths and thickness of layers and regions may be exaggerated for convenience. It will be understood that, when a first element is referred to as being “on” a second element, although the first element may be disposed on the second element so as to come into contact with the second element, a third element may be interposed between the first element and the second element.

(13) Here, terms such as, for example, “first” and “second” may be used to distinguish any one element with another element. However, the first element and the second element may be arbitrary named according to the convenience of those skilled in the art without departing the technical spirit of the present invention.

(14) The terms used in the specification of the present invention are merely used in order to describe particular embodiments, and are not intended to limit the scope of the present invention. For example, an element described in the singular form is intended to include a plurality of elements unless the context clearly indicates otherwise. In addition, in the specification of the present invention, it will be further understood that the terms “comprises” and “includes” specify the presence of stated features, integers, steps, operations, elements, components, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or combinations.

(15) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Embodiment

(16) FIG. 1 is a view schematically showing a display apparatus according to an embodiment of the present disclosure. FIG. 2 is an enlarged view showing a peripheral of a substrate hole in the display apparatus according to the embodiment of the present disclosure. FIG. 3A is a view showing a cross-section of a pixel in the display apparatus according to the embodiment of the present disclosure. FIG. 3B is a view showing a cross-section of a peripheral region of the substrate hole in the display apparatus according to the embodiment of the present disclosure. FIG. 4 is an enlarged view of K in FIG. 3B according to an embodiment of the present disclosure.

(17) Referring to FIGS. 1 to 4, the display apparatus according to the embodiment of the present disclosure may include a device substrate **100**. The device substrate **100** may include an insulating material. For example, the device substrate **100** may include glass or plastic.

(18) Gate lines GL and data lines DL may be disposed on the device substrate **100**. The data lines DL may intersect the gate lines GL. For example, the gate lines GL and the data lines DL may define a plurality of pixel regions PA.

(19) Each of the pixel regions PA may realize a specific color. For example, a light-emitting device **700** may be disposed in each pixel region PA. The light-emitting device **700** may emit light

displaying a specific color. For example, the light-emitting device **700** may include a first electrode **710**, a light-emitting layer **720** and a second electrode **730**, which are sequentially stacked.

(20) The first electrode **710** may include a conductive material. The first electrode **710** may include a metal having a relatively high reflectance. For example, the first electrode **710** may include a metal, such as aluminum (Al) and silver (Ag). The first electrode **710** may have a multi-layer structure. For example, the first electrode **710** may have a structure in which a reflective electrode formed of a metal is disposed between transparent electrodes formed of a transparent conductive material such as ITO and IZO.

(21) The light-emitting layer **720** may generate light having luminance corresponding to a voltage difference between the first electrode **710** and the second electrode **730**. For example, the light-emitting layer **720** may include an emission material. The emission material may include an organic material, an inorganic material or a hybrid material. For example, the display apparatus according to the embodiment of the present disclosure may be an organic light-emitting display apparatus including the light-emitting layer **720** of an organic material.

(22) The light-emitting layer **720** may have a multi-layer structure. For example, the light-emitting layer **720** may include an emission material layer (EML) **722** disposed between a first organic layer **721** and a second organic layer **723**. The first organic layer **721** and the second organic layer **723** may include at least one of a hole injection layer (HIL) and a hole transporting layer (HTL), an electron transporting layer (ETL) and an electron injection layer (EIL). Thus, in the display apparatus according to the embodiment of the present disclosure, the emission efficiency of the light-emitting device **700** may be improved.

(23) The second electrode **730** may include a conductive material. The second electrode **730** may include a material different from the first electrode **710**. For example, the second electrode **730** may be a transparent electrode formed of a transparent conductive material, such as ITO and IZO. Thus, in the display apparatus according to the embodiment of the present invention, the light generated from the light-emitting layer **720** of each pixel PA may emit outside through the second electrode **730**.

(24) A driving circuit D connected to the light-emitting device **700** may be disposed in each pixel region PA. The driving circuit D may supply a driving current corresponding to a gate signal applied by the gate lines GL and a data signal applied by the data line DL to the light-emitting device **700**. For example, the driving circuit D may include a first thin film transistor **200**, a second thin film transistor **300** and a storage capacitor **400**.

(25) The first thin film transistor **200** may turn on/off the second thin film transistor **300** according to the gate signal. The first thin film transistor **200** may include a first semiconductor pattern **210**, a first gate electrode **230**, a first source electrode **250** and a first drain electrode **260**. The first gate electrode **230** may overlap a portion of the first semiconductor pattern **210**. A first gate insulating layer **121** may be disposed between the first semiconductor pattern **210** and the first gate electrode **230**. The first source electrode **250** and the first drain electrode **260** may be connected to the first semiconductor pattern **210**.

(26) The first semiconductor pattern **210** may be disposed close to the device substrate **100**. The first semiconductor pattern **210** may include a semiconductor material. The first semiconductor pattern **210** may include a poly-silicon. For example, the first semiconductor pattern **210** may include a low-temperature poly silicon (LIPS). The first semiconductor pattern **210** may include a first channel region disposed between a first source region and a first drain region. The first source region and the first drain region may have a higher conductivity than the first channel region. For example, the first source region and the first drain region may have a higher content of conductive impurities than the first channel region.

(27) The first gate insulating layer **121** may be disposed on the first semiconductor pattern **210**. The first gate insulating layer **121** may extend beyond the first semiconductor pattern **210**. For example, a side surface of the first semiconductor pattern **210** may be covered by the first gate insulating

layer **121**. The first gate insulating layer **121** may include an insulating material. For example, the first gate insulating layer **121** may include a silicon oxide (SiO_x) or silicon nitride (SiN_x). The first gate insulating layer **121** may include a material having a high dielectric constant (High-K material). For example, the first gate insulating layer **121** may include hafnium oxide (HfO). The first gate insulating layer **121** may have a multi-layer structure.

(28) The first gate electrode **230** may be disposed on the first gate insulating layer **121**. For example, the first gate electrode **230** may overlap the channel region of the first semiconductor pattern **210**. The first gate electrode **230** may be insulated from the first semiconductor pattern **210** by the first gate insulating layer **121**. The first gate electrode **230** may include a conductive material. For example, the first gate electrode **230** may include a metal, such as aluminum (Al), chromium (Cr), copper (Cu), titanium (Ti), molybdenum (Mo) and tungsten (W). The first gate electrode **230** may be formed of a single layer or a multiple layer.

(29) The first source electrode **250** may be electrically connected to the first source region of the first semiconductor pattern **210**. For example, the first source electrode **250** may be in direct contact with a portion of the first source region. The first source electrode **250** may include a portion overlapping with the first source region. The first source electrode may be insulated from the first gate electrode **230**. For example, a lower interlayer insulating layer **120** may be disposed on the first gate insulating layer **121** and the first gate electrode **230**, and the first source electrode **250** may be disposed on the lower interlayer insulating layer **120**. For example, the lower interlayer insulating layer **120** may include silicon oxide (SiO_x). The lower interlayer insulating layer **120** may extend beyond the first semiconductor pattern **210**. For example, a side surface of the first gate electrode **230** may be in direct contact with the lower interlayer insulating layer **120**. The lower interlayer insulating layer **120** may include a first source contact hole partially exposing the first source region of the first semiconductor pattern **210**. The first source electrode **250** may be connected to the first source region of the first semiconductor pattern **210** in the first source contact hole.

(30) The first source electrode **250** may include a conductive material. For example, the first source electrode **250** may include a metal, such as aluminum (Al), titanium (Ti), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu) or alloy thereof. The first source electrode **250** may include a material different from the first gate electrode **230**.

(31) The first drain electrode **260** may be electrically connected to the first drain region of the first semiconductor pattern **210**. For example, the first drain electrode **260** may be in direct contact with a portion of the first drain region. The first drain electrode **260** may include a portion overlapping with the first drain region. The first drain electrode **260** may be insulated from the first gate electrode **230**. For example, the first drain electrode **260** may be disposed on the lower interlayer insulating layer **120**. The first drain electrode **260** may be spaced away from the first source electrode **250**. For example, the lower interlayer insulating layer **120** may include a first drain contact hole partially exposing the first drain region. The first drain electrode **260** may be connected to the first drain region in the first drain contact hole.

(32) The first drain electrode **260** may include a conductive material. For example, the first drain electrode **260** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu), or alloy thereof. The first drain electrode **260** may be formed of a single layer or a multiple layer. The first drain electrode **260** may include the same material as the first source electrode **250**. The first drain electrode **260** may include a material different from the first gate electrode **230**.

(33) The second thin film transistor **300** may generate a driving current corresponding to the data signal. The second thin film transistor **300** may include a second semiconductor pattern **310**, a second gate electrode **330**, a second source electrode **350** and a second drain electrode **360**. The second gate electrode **330** may overlap a portion of the second semiconductor pattern **310**. A second gate insulating layer **141** may be disposed between the second semiconductor pattern **310**

and the second gate electrode **330**. The second source electrode **350** and the second drain electrode **360** may be connected to the second semiconductor pattern **310**.

(34) The second semiconductor pattern **310** may include a material different from the first semiconductor pattern **210**. The second semiconductor pattern **310** may be an oxide semiconductor. For example, the second semiconductor pattern **310** may include a metal oxide, such as IGZO.

(35) The second semiconductor pattern **310** may be disposed on a layer different from the first semiconductor pattern **210**. For example, an intermediate buffer layer **140** may be disposed between the first semiconductor pattern **210** and the second semiconductor pattern **310**. The first semiconductor pattern **210** may be disposed between the device substrate **100** and the intermediate buffer layer **140**. The second semiconductor pattern **310** may be disposed on the intermediate buffer layer **140**.

(36) The intermediate buffer layer **140** may include silicon nitride (SiN_x) or silicon oxide (SiO_x). Referring to FIG. 3A, the intermediate buffer layer **140** is illustrated as a single layer, but is not limited thereto. For example, the intermediate buffer layer **140** may be formed of a multiple layer in which a layer including silicon nitride (SiN_x) and a layer including silicon oxide (SiO_x) are stacked.

(37) The second semiconductor pattern **310** may have the same configuration as the first semiconductor pattern **210**. For example, the second semiconductor pattern **310** may include a second channel region disposed between a second source region and a second drain region. An electrical conductivity of the second source region and an electrical conductivity of the second drain region may be higher than an electrical conductivity of the second channel region. For example, the second source region and the second drain region may be a conductorized region.

(38) The second gate insulating layer **141** may be disposed on the second semiconductor pattern **310**. For example, the second gate insulating layer **141** may overlap the second channel region of the second semiconductor pattern **310**. The second source region and the second drain region of the second semiconductor pattern **310** may be disposed outside the second gate insulating layer **141**. The second gate insulating layer **141** may include an insulating material. For example, the second gate insulating layer **320** may include silicon oxide (SiO_x) or silicon nitride (SiN_x). The second gate insulating layer **141** may include a material having a high dielectric constant (High-K material). For example, the second gate insulating layer **141** may include hafnium oxide (HfO). The second gate insulating layer **141** may have a multi-layer structure. The second gate insulating layer **141** may have the same stacked structure as the first gate insulating layer **121**.

(39) The second gate electrode **330** may be disposed on the second gate insulating layer **141**. The second gate electrode **330** may overlap the second channel region of the second semiconductor pattern **310**. The second gate electrode **330** may be insulated from the second semiconductor pattern **310** by the second gate insulating layer **141**. The second channel region of the second semiconductor pattern **310** may have an electrical conductivity corresponding to a voltage applied to the second gate electrode **330**. For example, the second channel region of the second semiconductor pattern **310** may be a semiconductor region.

(40) The second gate electrode **330** may include a conductive material. For example, the second gate electrode **330** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu). The second gate electrode **330** may have a single layer or a multiple layer. The second gate electrode **330** may include the same material as the first gate electrode **230**.

(41) The second source electrode **350** may be electrically connected to the second source region of the second semiconductor pattern **310**. For example, the second source electrode **350** may be in direct contact with a portion of the second source region. The second source electrode **350** may be insulated from the second gate electrode **330**. For example an upper interlayer insulating layer **150** may be disposed on the second semiconductor pattern **310** and the second gate electrode **330**, and the second source electrode **350** may be disposed on the upper interlayer insulating layer **150**. The

upper interlayer insulating layer **150** may include an insulating material. For example, the upper interlayer insulating layer **150** may include silicon oxide (SiOx). The upper interlayer insulating layer **150** may extend beyond the second semiconductor pattern **310**. For example, a side surface of the second gate electrode **330** may be in direct contact with the upper interlayer insulating layer **150**. The upper interlayer insulating layer **150** may include a second source contact hole partially exposing the second source region of the second semiconductor pattern **310**. The second source electrode **350** may be connected to the second source region of the second semiconductor pattern **310** in the second source contact hole.

(42) The upper interlayer insulating layer **150** may include silicon nitride (SiNx) or silicon oxide (SiOx). Referring to FIG. 3A, the upper interlayer insulating layer **150** is illustrated as a single layer, but is not limited thereto. For example, the upper interlayer insulating layer **150** may have a multi-layer structure in which a layer formed of silicon nitride (SiNx) and a layer formed of silicon oxide (SiOx) are stacked.

(43) The second source electrode **350** may include a conductive material. For example, the second source electrode **350** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu). The second source electrode **350** may have a single layer or a multiple layer. The second source electrode **350** may include the same material as the first source electrode **250**.

(44) The second drain electrode **360** may be electrically connected to the second drain region of the second semiconductor pattern **310**. For example, the second drain electrode **360** may be in direct contact with a portion of the second drain region. The second drain electrode **360** may be insulated from the second gate electrode **330**. For example, the second drain electrode **360** may be disposed on the upper interlayer insulating layer **150**. The upper interlayer insulating layer **150** may include a second drain contact hole partially exposing the second drain region of the second semiconductor pattern **310**. The second drain electrode **360** may be connected to the second drain region of the second semiconductor pattern **310** in the second drain contact hole.

(45) The second drain electrode **360** may include a conductive material. For example, the second drain electrode **360** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu). The second drain electrode **360** may have a single layer or a multiple layer. The second drain electrode **360** may include the same material as the second source electrode **350**. The second drain electrode **360** may include a material different from the second gate electrode **330**.

(46) The storage capacitor **400** may be formed by a process of forming the first thin film transistor **200** and the second thin film transistor **300**. For example, the storage capacitor **400** may include a first storage electrode **410** disposed on the same layer as the first gate electrode **230**, and a second storage electrode **420** above the first storage electrode **410**.

(47) The first storage electrode **410** may include a conductive material. The first storage electrode **410** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu). The first storage electrode **410** may have a single layer or a multiple layer. The first storage electrode **410** may include the same material as the first gate electrode **230**.

(48) The second storage electrode **420** may include a conductive material. The second storage electrode **420** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu). The second storage electrode **420** may have a single layer or a multiple layer. The lower interlayer insulating layer **120** may extend between the first storage electrode **410** and the second storage electrode **420**. The second storage electrode **420** may be disposed on the lower interlayer insulating layer **120**. The first source electrode **250** and the first drain electrode **260** may be disposed a layer different from the second storage electrode **420**. For example, an intermediate interlayer insulating layer **130** may be disposed on the second storage electrode **420**, and the lower interlayer insulating layer **120** and the intermediate interlayer insulating layer **130** may be stacked between the first gate electrode **230** and the first source

electrodes **250**, and between the first gate electrode **230** and the first drain electrode **260**. The second storage electrode **420** may include a material different from the first source electrode **250** and the first drain electrode **260**.

(49) The intermediate interlayer insulating layer **130** may include an insulating material. For example, the intermediate interlayer insulating layer **130** may include silicon oxide (SiOx) and/or silicon nitride (SiNx). The intermediate interlayer insulating layer **130** may have a single layer or a multiple layer. For example, the intermediate interlayer insulating layer **130** may have a multi-layer structure in which an insulating layer formed of silicon oxide (SiOx) and an insulating layer formed of silicon nitride (SiNx) are stacked.

(50) The second storage electrode **420** may be electrically connected to the second drain electrode **360** of the second thin film transistor **300**. For example, a storage connecting electrode **450** penetrating the intermediate interlayer insulating layer **130** may be disposed on the intermediate interlayer insulating layer **130**. The storage connecting electrode **450** may be disposed on the same layer as the first source electrode **250** and the first drain electrode **260**. For example, the storage connecting electrode **450** may include the same material as the first source electrode **250** and the first drain electrode **260**. The second drain electrode **360** may be connected to the storage connecting electrode **450** by penetrating the intermediate buffer layer **140** and the upper interlayer insulating layer **150**.

(51) A light-blocking electrode **500** may be disposed between the lower interlayer insulating layer **120** and the intermediate interlayer insulating layer **130**. The light-blocking electrode **500** may prevent the characteristics change of the second semiconductor pattern **310** due to the external light. For example, the light-blocking electrode **500** may include a metal. The light-blocking electrode **500** may include the same material as the second storage electrode **420**. For example, the light-blocking electrode **500** may be disposed on the same layer as the second storage electrode **420**.

(52) A device buffer layer **110** may be disposed between the device substrate **100** and the driving circuit D of each pixel region PA. The device buffer layer **110** may prevent pollution from the device substrate **100** during a process of forming the driving circuits D. For example, the device buffer layer **110** may extend between the device substrate **100** and the first semiconductor pattern **210** of each pixel region PA. The device buffer layer **110** may include an insulating material. For example, the device buffer layer **110** may include silicon oxide (SiOx) and/or silicon nitride (SiNx). The device buffer layer **110** may have a multi-layer structure. For example, the device buffer layer **110** may have a stacked structure of a first buffer layer **111** and a second buffer layer **112** including a material different from the first buffer layer **111**.

(53) A first over-coat layer **160** and a second over-coat layer **170** may be sequentially stacked between the driving circuit D and the light-emitting device **700** of each pixel region PA. The first over-coat layer **160** may remove a thickness difference due to the driving circuit D of each pixel region PA. The second over-coat layer **170** may be disposed on the first over-coat layer **160**. A surface of the second over-coat layer **170** toward the light-emitting device **700** of each pixel region PA may be a flat surface. The first over-coat layer **160** and the second over-coat layer **170** may include an insulating material. The first over-coat layer **160** and the second over-coat layer **170** may include a material different from the upper interlayer insulating layer **150**. For example, the first over-coat layer **160** and the second over-coat layer **170** may include an organic insulating material. The second over-coat layer **170** may include a material different from the first over-coat layer **160**, but is not limited thereto.

(54) The light-emitting device **700** of each pixel region PA may be electrically connected to the second thin film transistor **300** of the corresponding pixel region PA. For example, the first electrode **710** of each pixel region PA may be electrically connected to the corresponding second drain electrode **360** by penetrating the first over-coat layer **160** and the second over-coat layer **170**. The first electrode **710** of each pixel region PA may be electrically connected to the corresponding

second drain electrode **360** by an intermediate electrode **610**. For example, the intermediate electrode **610** may be disposed between the first over-coat layer **160** and the second over-coat layer **170**. The intermediate electrode **610** may be connected to the second drain electrode **360** by penetrating the first over-coat layer **160**, and the first electrode **710** may be connected to the intermediate electrode **610** by penetrating the second over-coat layer **170**.

(55) The intermediate electrode **610** may include a conductive material. For example, the intermediate electrode **610** may include a metal, such as aluminum (Al), chromium (Cr), titanium (Ti), molybdenum (Mo), tungsten (W) and copper (Cu), or alloy thereof. The intermediate electrode **620** may have a single layer or a multiple layer.

(56) The light-emitting device **700** of each pixel region PA may be independently operated. For example, the first electrode **710** of each pixel region PA may be insulated from the first electrode **710** of adjacent pixel region PA. An edge of each first electrode **710** may be covered by a bank insulating layer **180**. The bank insulating layer **180** may be disposed on the second over-coat layer **170**. The light-emitting layer **720** and the second electrode **730** of each pixel region PA may be stacked on a portion of the corresponding first electrode **710** exposed by the bank insulating layer **180**. The bank insulating layer **180** may include an insulating material. For example, the bank insulating layer **180** may include an organic insulating material. The bank insulating layer **180** may include a material different from the second over-coat layer **170**.

(57) At least a portion of the light-emitting layer **720** of each pixel region PA may extend onto the bank insulating layer **180**. For example, the first organic layer **721** and the second organic layer **723** of each pixel region PA may be connected to the first organic layer **721** and the second organic layer **723** of adjacent pixel region PA. The emission material layer **722** of each pixel region PA may be spaced away from the emission material layer **722** of adjacent pixel region PA. For example, the emission material layer **722** of each pixel region PA may be formed by a deposition process using a fine metal mask (FMM). A spacer **800** may be disposed on the bank insulating layer **180**. The spacer **800** may prevent damage of adjacent emission material layer **722** and/or the bank insulating layer **180** due to the fine metal mask. For example, the end of each emission material layer **722** may be disposed on a surface of the bank insulating layer **180** which is disposed outside the spacer **800**. The spacer **800** may include an insulating material.

(58) The second electrode **730** of each pixel region PA may extend onto the bank insulating layer **180**. For example, the second electrode **730** of each pixel region PA may be connected to the second electrode **730** of adjacent pixel region PA.

(59) A substrate hole CH may be formed in the device substrate **100**. The substrate hole CH may be disposed among the pixel regions PA. For example, the substrate hole CH may be formed among the light-emitting devices **700**. The device substrate **100** may include a hole peripheral region HA including a region in which the substrate hole CH is formed. The light-emitting devices **700** may be disposed outside the hole peripheral region HA. The gate lines GL and the data lines DL may be bypassed along an edge of the substrate hole CH in the hole peripheral region HA.

(60) The hole peripheral region HA may include a penetrating region CA in which the substrate hole CH is formed, a separating region BA surrounding the penetrating region CA, and a dam region DA disposed outside the separating region BA. The separating region BA may be disposed between the penetrating region CA and the dam region DA. For example, the separating region BA may be disposed between the substrate hole CH and the light-emitting devices **700**.

(61) A separating device **911**, **912** and **913** may be disposed in the separating region BA. The separating device **911**, **912** and **913** may be in direct contact with the device substrate **100**. For example, an insulating layer formed in the separating region BA may be removed. Thus, in the display apparatus according to the embodiment of the present disclosure, a crack which occurs in a process of forming the substrate hole CH may be blocked in the separating region BA. Therefore, in the display apparatus according to the embodiment of the present disclosure, the permeation of

moisture due to the process of forming the substrate hole CH may be prevented.

(62) The separating device **911**, **912** and **913** may have a structure in which a metal pattern **912m** and an insulating cap **912c** are stacked. The metal pattern **912m** may have a smaller width than the insulating cap **912c**. For example, an under-cut UC may be formed between the device substrate **100** and the insulating cap **912c**. Thus, in the display apparatus according to the embodiment of the present disclosure, the organic and/or inorganic layer deposited after the formation of the separating device **911**, **912** and **913** may be separated by the separating device **911**, **912** and **913**. For example, the first organic layer **721**, the second organic layer **723** and the second electrode **730** may be partially cut off by the separating device **911**, **912** and **913**.

(63) The separating device **911**, **912** and **913** may have various shapes. For example, at least one separating device **912** may have a shape in which the under-cut is formed at a left lower end and a right lower end of the insulating cap **912c**. And, at least one separating device **911** and **913** may have a shape in which the insulating cap extends beyond one side of the metal pattern. Thus, the display apparatus according to the embodiment of the present disclosure may effectively block the permeation of the external moisture using the separating device **911**, **912** and **913** having various shapes.

(64) The metal pattern **912m** may include the same material as the intermediate electrode **610**. For example, the metal pattern **912m** may have the same stacked structure as the intermediate electrode **610**.

(65) A dam **921**, **922** and **923** may be disposed in the dam region DA. At least one insulating layer **121**, **130**, **140** and **150** may be stacked between the device substrate **100** and the dam **921**, **922** and **923**. The dam **921**, **922** and **923** may have the same structure as the separating device **911**, **912** and **913**. For example, at least one dam **922** may have a shape in which a metal pattern and an insulating cap are sequentially stacked, and an under-cut is formed at a left lower end and a right lower end of the insulating cap. And, at least one dam **921** and **923** may have a shape in which a metal pattern and an insulating cap are sequentially stacked, and the insulating cap extends beyond one side of the metal pattern. Thus, the display apparatus according to the embodiment of the present disclosure may effectively block the permeation of the external moisture using the separating device **911**, **912** and **913**, and the dam **921**, **922** and **923** which have various shapes.

(66) FIGS. 5A to 10A and 5B to 10B are views sequentially showing a method of forming the display apparatus according to the embodiment of the present disclosure.

(67) A method of forming the display device according to the embodiment of the present disclosure will be described with reference to FIGS. 3A, 3B, 5A to 10A and 5B to 10B. First, as shown in FIGS. 5A and 5B, the method of forming the display apparatus according to the embodiment of the present disclosure may include a step of forming a driving circuit D on each pixel region PA of the device substrate **100**, a step of forming the first over-coat layer **160** covering the driving circuits D, and a step of removing an insulating layer which is disposed on the separating region BA and the penetrating region CA.

(68) The step of removing the insulating layer in the separating region BA and the penetrating region CA may be performed relatively gently between the dam region DA and the separating region BA due to a difference in density. For example, insulating layers **121**, **130**, **140** and **150** stacked on the dam region DA may include a side surface toward the penetrating region CA, and the side surface of the insulating layers **121**, **130**, **140** and **150** toward the penetrating region CA may have a gentle inclination.

(69) As shown in FIGS. 6A and 6B, the method of forming the display apparatus according to the embodiment of the present disclosure may include a step of forming the intermediate electrode **610** on the first over-coat layer **160**, a step of forming a first preliminary pattern **910** on the separating region BA, and a step of forming a second preliminary pattern **920** on the dam region DA.

(70) The first preliminary pattern **910** and the second preliminary pattern **920** may be formed simultaneously with the intermediate electrode **610**. For example, a step of forming the

intermediate electrode **610**, the first preliminary pattern **910** and the second preliminary pattern **920** may include a step of forming a conductive material layer on the first over-coat layer **160**, and a step of patterning the conductive material layer. Thus, in the method of forming the display apparatus according to the embodiment of the present invention, the intermediate electrode **610**, the first preliminary pattern **910** and the second preliminary pattern **920** may be formed of the same material.

(71) As shown in FIGS. **7A** and **7B**, the method of forming the display apparatus according to the embodiment of the present disclosure may include a step of forming the second over-coat layer **170** on the intermediate electrode **610**, the first preliminary pattern **910** and the second preliminary pattern **920**, a step of forming a first preliminary cap **915** on the first preliminary pattern **910**, a step of forming a second preliminary cap **925** on the second preliminary pattern **920**, and a step of forming the first electrode **710** connected to the intermediate electrode **610** on the second over-coat layer **170**.

(72) The first preliminary cap **915** and the second preliminary cap **925** may be formed by using the second over-coat layer **170**. For example, the step of forming the first preliminary cap **915** and the second preliminary cap **925** may include a step of partially etching the second over-coat layer **170** on the separating region BA and the dam region DA. The second preliminary cap **925** may be formed of the same material as the first preliminary cap **915**.

(73) The first preliminary cap **915** may partially expose the first preliminary pattern **910**. The second preliminary cap **925** may partially expose the second preliminary pattern **920**. For example, the location of the first preliminary cap **915** based on the first preliminary pattern **910** may correspond to the location of the second preliminary cap **925** based on the second preliminary pattern **920**.

(74) As shown in FIGS. **8A** and **8B**, the method of forming the display apparatus according to the embodiment of the present disclosure may include a step of forming the bank insulating layer **180**, the spacer **800**, a first upper cap **917** and a second upper cap **927** on the device substrate **100** in which the first electrode **710** of each pixel region PA is formed.

(75) The bank insulating layer **180** may cover the edge of the first electrode **710** of each pixel region PA. The spacer **800** may be formed on the bank insulating layer **180**. The first upper cap **917** may be formed on the first preliminary cap **915**. The second upper cap **927** may be formed on the second preliminary cap **925**.

(76) The first upper cap **917** and the second upper cap **927** may be formed by using a process of forming the bank insulating layer **180** and the spacer **800**. For example, the step of forming first upper cap **917** and the second upper cap **927** may include a step of stacking insulating layers for forming the bank insulating layer **180** and the spacer **800**, and a step of patterning the insulating layers. The second upper cap **927** may be formed of the same material as the first upper cap **917**. For example, the first upper cap **917** and the second upper cap **927** may be formed simultaneously with the bank insulating layer **180** and/or the spacer **800**.

(77) As shown in FIGS. **9A** and **9B**, the method of forming the display apparatus according to the embodiment of the present disclosure may include a step of removing a portion of the first preliminary pattern **910** exposed by the first preliminary cap **915** and the first upper cap **917**, and a step of removing a portion of the second preliminary pattern **920** exposed by the second preliminary cap **925** and the second upper cap **927**.

(78) The step of removing a portion of the second preliminary pattern **920** may be performed simultaneously with the step of removing a portion of the first preliminary pattern **910**. For example, the step of removing a portion of the second preliminary pattern **920** and a portion of the first preliminary pattern **910** may include a step of etching the first preliminary pattern **910** and the second preliminary pattern **920** using the first upper cap **917** and the second upper cap **927** as an etching mask. A first preliminary metal pattern **910a** overlapping with the first preliminary cap **915** and a second preliminary metal pattern **920a** overlapping with the second preliminary cap **925** may

be formed by the process of etching the first preliminary pattern **910** and the second preliminary pattern **920**.

(79) As shown in FIGS. **10A** and **10B**, the method of forming the display apparatus according to the display apparatus according to the embodiment of the present disclosure may include a step of forming the separating device **911**, **912** and **913** on the separating region BA and the dam **921**, **922** and **923** on the dam region DA.

(80) The step of forming the separating device **911**, **912** and **913**, and the dam **921**, **922** and **923** may include a step of reducing a width of the first preliminary metal pattern **910a** and the second preliminary metal pattern **920a**. Thus, in the display apparatus according to the embodiment of the present disclosure, the separating device **911**, **912** and **913** and the dam **921**, **922** and **923** may include the metal pattern and the insulating cap having a larger width than the metal pattern, which are stacked. That is, in the display apparatus according to the embodiment of the present disclosure, each of the separating device **911**, **912** and **913** and the dam **921**, **922** and **923** may include the under-cut due to a difference in width.

(81) As shown in FIGS. **11A** and **11B**, the method of forming the display apparatus according to the embodiment of the present disclosure may include a step of forming the light-emitting layer **720** and the second electrode **730** of each light-emitting device **700** on the device substrate **100** in which the separating device **911**, **912** and **913** and the dam **921**, **922** and **923** are formed, and a step of forming the substrate hole CH in the penetrating region CA.

(82) Accordingly, in the display apparatus according to the embodiment of the present disclosure and the method of forming the same, the insulating layers stacked on the separating region BA between the penetrating region CA in which the substrate hole CH is formed, and the pixel regions PA may be completely removed, so that the separating device **911**, **912** and **913** formed on the separating region BA by a subsequent process may be in contact with the device substrate **100**. Thus, in the display apparatus according to the embodiment of the present disclosure and the method of forming the same, the crack which occurs in the inorganic insulating layer by a process of forming the substrate hole may not propagate to the pixel regions PA. Thereby, in the display apparatus according to the embodiment of the present disclosure and the method of forming the same, the permeation of the external moisture due to a process of forming the substrate hole may be blocked, and the damage of the light-emitting devices due to the external moisture may be prevented.

(83) And, in the display apparatus according to the embodiment of the present disclosure and the method of forming the same, the separating device **911**, **912** and **913** may be formed simultaneously with the dam **921**, **922** and **923**. Therefore, in the display apparatus according to the embodiment of the present disclosure and the method of forming the same, process efficiency may be improved.

(84) The display apparatus according to the embodiment of the present disclosure is described that the device substrate **100** is a single layer. However, in the display apparatus according to another embodiment of the present disclosure, the device substrate **100** may have a multi-layer structure. For example, in the display apparatus according to another embodiment of the present disclosure, the device substrate **100** may have a structure in which an insulating layer **102** may be disposed between the first substrate layer **101** and the second substrate layer **103**, as shown in FIGS. **11A** and **11B**. The second substrate layer **103** may include the same material as the first substrate layer **101**. For example, the first substrate layer **101** and the second substrate layer **103** may include plastic. The insulating layer **102** may include an insulating material. Thus, in the display apparatus according to another embodiment of the present disclosure, the device substrate **100** may include a recess groove **100g** disposed adjacent the separating device **911**, **912** and **913**.

(85) FIGS. **13A** and **13B** are views sequentially showing a method of forming the display apparatus according to another embodiment of the present disclosure.

(86) As shown in FIGS. **13A** and **13B**, in the method of forming the display apparatus according to

another embodiment of the present disclosure, an over-etching may occur in a process of forming the first preliminary metal pattern **910a**. Thus, in the method of forming the display apparatus according to another embodiment of the present disclosure, a portion of the device substrate **100** exposed by the first preliminary cap **915** may be etched, so that the recess groove **100g** may be formed.

(87) The step of forming the first preliminary metal pattern **910a** may be performed simultaneously with the step of forming the second preliminary metal pattern **920a**. For example, a portion of the upper interlayer insulating layer **150** exposed by the second preliminary cap **925** may be over-etched in the step of forming the second preliminary metal pattern **920a**.

(88) That is, the display apparatus according to another embodiment of the present disclosure may include the recess groove **100g** in a portion of the device substrate **100** which is formed by the step of forming the separating device **911**, **912** and **913**. Thus, in the display apparatus according to another embodiment of the present disclosure, the moving path of the external moisture permeated through the substrate hole CH formed in the penetrating hole CA may be increased. Thereby, in the display apparatus according to another embodiment of the present disclosure, the deterioration of the light-emitting devices due to the permeation of the external moisture may be effectively prevented.

(89) In the result, the display apparatus according to the embodiments of the present invention may include the separating device between the substrate hole and the light-emitting devices, wherein the separating device may be in contact with the device substrate. Thus, in the display apparatus according to the embodiments of the present disclosure, the crack which occurs in the inorganic insulating layer in a process of forming the substrate hole may be blocked by the separating device. Thereby, in the display apparatus according to the embodiments of the present disclosure, the reliability of the light-emitting devices may be improved.

Claims

1. A display apparatus comprising: a substrate hole penetrating a device substrate; a light-emitting device on the device substrate, the light-emitting device spaced away from the substrate hole; a plurality of inorganic insulating layers between the device substrate and the light-emitting device; a separating device disposed laterally between the plurality of inorganic insulating layers and the substrate hole; a dam on the plurality of inorganic insulating layers, the dam disposed laterally between the light-emitting device and the separating device, the dam including a first metal pattern and a first insulating cap disposed on the first metal pattern; and a first over-coat layer between the plurality of inorganic insulating layers and the light-emitting device, wherein the dam, the separating device and the substrate hole are laterally spaced apart from the first over-coat layer, and a driving circuit a transistor including at least one thin film transistor and an intermediate electrode which electrically connects the light-emitting device to the driving circuit through a first contact hole penetrating the first over-coat layer, wherein the plurality of inorganic insulating layers includes at least one gate insulating layer of the at least one transistor, wherein the separating device includes a second metal pattern being in direct contact with the device substrate, and a second insulating cap disposed on the second metal pattern, wherein a side surface of each of the plurality of inorganic insulating layers toward the separating device has a gentler slope than a side surface of the separating device and a side surface of the dam, such that an angle between a side surface of the plurality of inorganic insulating layers toward the separating device and an upper surface of the device substrate is smaller than an angle between a side surface of the first insulating cap of the dam toward the separating device and the upper surface of the device substrate and an angle between a side surface of the second insulating cap of the separating device toward the dam and the upper surface of the device substrate, and wherein the first metal pattern and the second metal pattern are patterned from a same layer as the intermediate electrode.

2. The display apparatus according to claim 1, wherein the device substrate includes a recess groove disposed adjacent the separating device, and wherein the separating device is spaced apart from the recess groove of the device substrate.
3. The display apparatus according to claim 2, wherein the recess groove of the device substrate has a thickness thinner than a portion of the device substrate overlapping with the separating device.
4. The display apparatus according to claim 1, wherein the first metal pattern overlaps with the first insulating cap and has a smaller width than the first insulating cap, and wherein the second metal pattern overlaps with the second insulating cap and has a smaller width than the second insulating cap.
5. The display apparatus according to claim 4, wherein at least one organic layer of the light-emitting device are cut off by the separating device and the dam.
6. The display apparatus according to claim 4, wherein the second metal pattern includes a same material as the first metal pattern, and wherein the second insulating cap includes a same material as the first insulating cap.
7. The display apparatus according to claim 6, wherein the dam has a same stacked structure as the separating device.
8. The display apparatus according to claim 1, wherein the side surface of the plurality of inorganic insulating layers toward the separating device is disposed between the dam and the separating device.
9. The display apparatus according to claim 1, wherein the first metal pattern is in contact with the plurality of inorganic insulating layers.
10. The display apparatus according to claim 1, further comprising: second over-coat layer disposed between the first over-coat layer and the light-emitting device; and a connecting electrode disposed between the first over-coat layer and the second over-coat layer, the connecting electrode connected to the light-emitting device through a contact hole penetrating the second over-coat layer, wherein the first metal pattern and the second metal pattern include a same material as the connecting electrode.
11. The display apparatus according to claim 10, wherein the first insulating cap and the second insulating cap include a same material as the second over-coat layer covering the connecting electrode.
12. The display apparatus according to claim 1, further comprising a first thin film transistor and a second thin film transistor disposed between the device substrate and the first over-coat layer, wherein a second semiconductor pattern of the second thin film transistor is disposed on a different layer from a first semiconductor pattern of the first thin film transistor, and wherein the second semiconductor pattern includes a different material from the first semiconductor pattern.
13. The display apparatus according to claim 12, wherein the first semiconductor pattern includes a low-temperature poly silicon (LTPS), and wherein the second semiconductor pattern includes an oxide semiconductor.
14. The display apparatus according to claim 12, wherein the plurality of inorganic insulating layers includes a first gate insulating layer, a second gate insulating layer on the first gate insulating layer, and an intermediate interlayer insulating layer, wherein the first gate insulating layer is disposed between the first semiconductor pattern and a first gate electrode of the first thin film transistor, wherein the second gate insulating layer is disposed between the second semiconductor pattern and a second gate electrode of the second thin film transistor, and wherein both ends of the first gate insulating layer or the second gate insulating layer are disposed on the second semiconductor pattern.
15. The display apparatus according to claim 12, further comprising a storage capacitor disposed between the device substrate and the first over-coat layer, wherein the storage capacitor includes a storage electrodes disposed on a same layer as a first gate electrode of the first thin film transistor.
16. The display apparatus according to claim 15, further comprising a light-blocking electrode

disposed between the device substrate and the second semiconductor pattern of the second thin film transistor, wherein the storage capacitor includes a storage electrodes disposed on a same layer as the light-blocking electrode.
